



IILM UNIVERSITY, GREATER NOIDA

Department of Computer Science & Engineering

INTERNSHIP REPORT

Google AI & Machine Learning Virtual Internship

Student Name	Deepak Kumar
Enrollment Number	2410031218
Program & Semester	B.Tech CSE • Semester IV
Institute	IILM University, Greater Noida
Internship Domain	Google AI-ML
Organization	Google for Developers / EduSkills
Internship Duration	10 Weeks (June – August 2026)
Academic Session	2026

1. Internship Overview

The Google AI & Machine Learning Virtual Internship is a 10-Week AICTE – EduSkills Virtual Internship Program, delivered in partnership with Google for Developers. The program is designed to build industry-relevant Artificial Intelligence and Machine Learning skills through structured, hands-on learning combined with practical project exposure.

Internship Duration

10 Weeks — June to August 2026, following a week-wise structured learning plan with weekly assessments, project submissions, and a final assessment test.

Focus Areas

- Machine Learning
- TensorFlow
- Neural Networks
- Object Detection
- Product Image Search

Purpose of the Internship

To strengthen employability and confidence by combining structured learning with practical, project-based exposure to real-world AI/ML applications, enabling students to demonstrate industry-relevant skills upon completion.

2. Internship Curriculum

The internship followed a week-wise structured curriculum spanning four core learning modules, culminating in an assessment and final credential validation.

Module 1: Program Neural Networks with TensorFlow

- Demystified machine learning from first principles
- Built convolutional neural networks (CNNs)
- Solved common computer-vision recognition and classification tasks

Module 2: Get Started with Object Detection

- Learned the basics of object detection
- Integrated a pretrained object detector into a mobile app

Module 3: Go Further with Object Detection

- Trained custom object-detection models
- Used TensorFlow Lite and the TensorFlow Lite Model Maker library

Module 4: Get Started with Product Image Search

- Built a product image search feature for a mobile app

- Detected objects in static images and live camera feeds on Android

3. Learning Objectives

- Understand Machine Learning Fundamentals
- Learn Data Preparation & Feature Engineering
- Train and Evaluate ML Models
- Learn TensorFlow Basics
- Understand Neural Networks
- Learn Object Detection Techniques
- Apply AI Concepts to Real-World Problems

4. AI & ML Technology Stack

Over the course of the internship, the following tools and technologies were used and learned:

Python	TensorFlow	TensorFlow Lite
Neural Networks	CNN	Computer Vision
Object Detection	ML Kit	Product Image Search

5. Learning Methodology

The internship followed a progressive, step-by-step learning path:

1. Learn AI & ML Fundamentals
2. Understand TensorFlow
3. Build Neural Networks
4. Train Models
5. Apply Object Detection
6. Deploy Models
7. Implement Product Image Search

6. Development Workflow

The end-to-end development pipeline followed during project work was as follows:

**Data Collection → Data Preparation → Feature Engineering → Model Training → Model Evaluation →
Object Detection → Deployment → AI Application**

7. Skills & Learning Outcomes

- Machine Learning Concepts

- TensorFlow Development
- Neural Networks
- Computer Vision
- Object Detection
- Model Evaluation
- AI Application Development

Overall Outcome: Strong foundation in Artificial Intelligence and Machine Learning.

8. Conclusion

- Successfully completed the Google AI & ML Internship
- Gained practical exposure to AI technologies
- Learned TensorFlow and Neural Networks
- Understood Object Detection and Product Image Search
- Built a strong foundation for future AI/ML projects

Appendix A: Internship Offer Letter

The internship offer letter issued by AICTE – EduSkills, confirming enrollment in the Google AI-ML Virtual Internship program, is reproduced below.

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Ref No: C15Y202602M021351239

Internship Offer Letter

Student Name	: Deepak Kumar
Stream	: B.Tech
Branch	: CSE
Institute Name	: IILM University, Greater Noida
Internship Domain	: Google AI-ML
Internship Duration	: June - August 2026
AICTE Student ID	: STU69988986a6e9a1771604358

We are pleased to inform you that you have been selected for the 10-Week AICTE – EduSkills Virtual Internship Program.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a week-wise structured learning plan.
- You will be required to complete weekly assessments to track your progress and understanding.
- Certain modules will require submission of project documentation and assignments.
- In the final week, you must appear for a Final Assessment Test.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a Virtual Internship Certificate.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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The week wise Structure of the Internship is as Follows:

Week	Module	Description
week 1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from the first principles in the new programming paradigm to creating convolutional neural networks for advanced image recognition and classification that solve common computer-vision problems.
week 2	Get started with object detection	Basics of object detection and how to integrate a pretrained object detector into your mobile app.
week 3	Go further with object detection	Train your own custom object-detection models using TensorFlow Lite and the TensorFlow Lite Model Maker library, and build on all the skills you gained in the Get started with object detection pathway.
week 4	Get started with product image search	Build a product image search feature for a mobile app using on-device Object Detection.
week 5	Get started with product image search	Detect objects in static images and live camera feeds on Android to build a visual product search.
week 6	Go further with product image search	Build and call a product image search backend from a mobile app, enhancing the product search feature you built in the Get started with product image search pathway.
week 7	Go further with image classification	Build custom image-classification models and improve the skills you gained in the Get started with image classification pathway.
week 8	Go further with image classification	Create a custom image classifier model and integrate it into your app.
week 9	Assessment	Assessment
week 10	Final Credential Validation	Final Credential Validation

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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Appendix B: Internship Completion Certificate

The Certificate of Virtual Internship, awarded upon successful completion of the 10-week AI-ML Virtual Internship, is reproduced below.



Certificate ID: 4de1ba3f1a15ca48ea98 | Student ID: STU69988986a6e9a1771604358